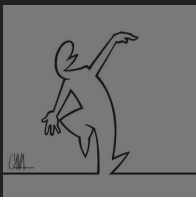


Reinforcement learning

why and how?



Nima H. Siboni
<https://github.com/nima-siboni>

Intro to myself

Feature position:

Senior RL engineer in Juna.ai

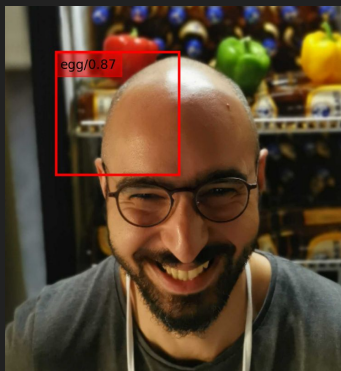
Former positions:

Senior RL research engineer in InstaDeep/BNT
Head of Machine Learning in DeepMetis

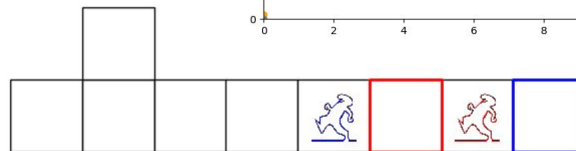
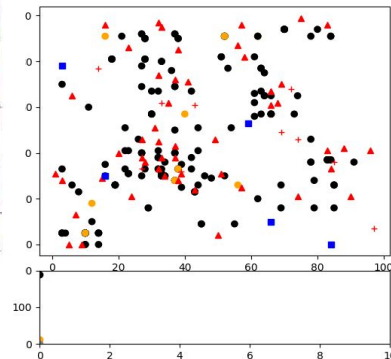
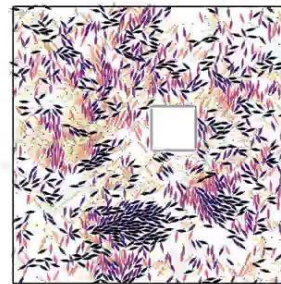
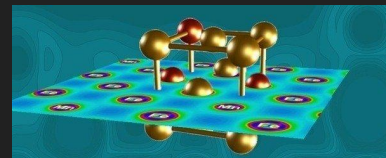
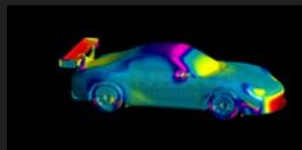
Other Affiliations:

- Max-Planck-Institute (guest researcher)
- AI-grid mentor

Short CV (in pictures)



- B.Sc. Mechanical Engineer
- M.Sc. Simulation Science
- PhD in Simulation of Complex Systems
- Postdoc in Simulation of Complex Fluids
- Data Science and Machine learning training



Course outline

- Introduction

What sort of problems you can solve with it? How is it new to you?

- What is Machine Learning?
- Different tasks in ML
- What is Reinforcement, what is Learning?
- Some examples
- Course disclaimer

- RL problem formulation

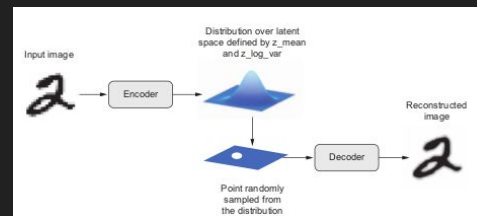
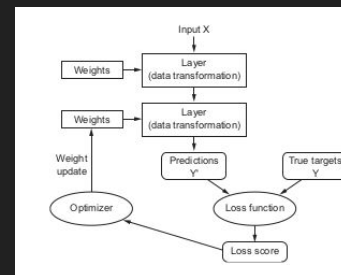
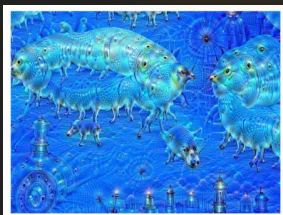
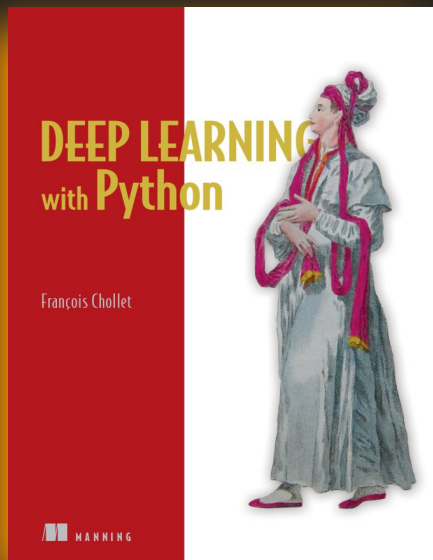
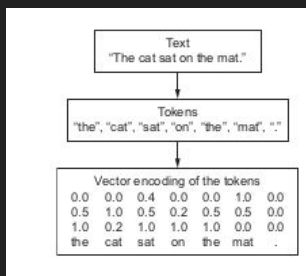
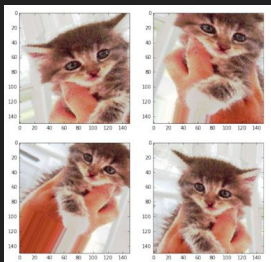
Lots of new terms to be defined and connected to each other

- RL's basic ingredients
- RL's problem formulation
- Exercise
- Anatomy of a RL solution

- Solution of a RL problem

Zoo of different methods

Course disclaimer!



What is Machine Learning?

Machine Learning: leveraging *data* to perform *tasks*

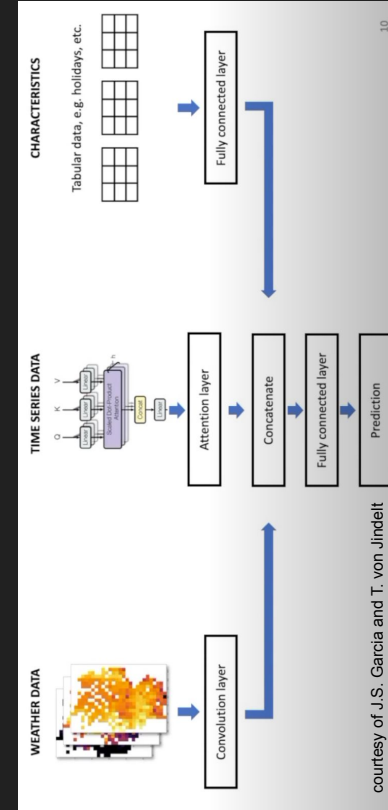
Task

Predicting the consumption and generation of renewable energy

Data

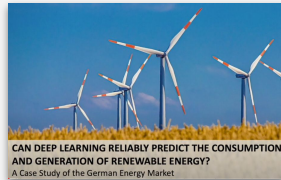
- What should prediction be based on?
 - Weather: wind, temperature, time of sun, pressure,
 - Features: day, week, month, holidays
 - Time series data
- What should the prediction replicate?
 - History
 - Expertise

The knowledge
to be *learned*

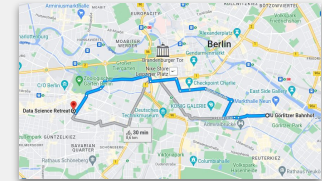


Different Tasks in ML

Machine Learning: leveraging *data* to perform *tasks*



- **"A" Regression or "A" Classification**



- **Sequential Decision Making**

What is Reinforcement, What is Learning?

Sequential Decision Making

- Your goal is to find an **optimal** *behavior*
- A *sequence* of decisions
- Each decision *affects* future decisions
- *Delayed* consequences

RL approach in a nutshell

Learning from **trials** (successes and failures) to optimize a decision making process

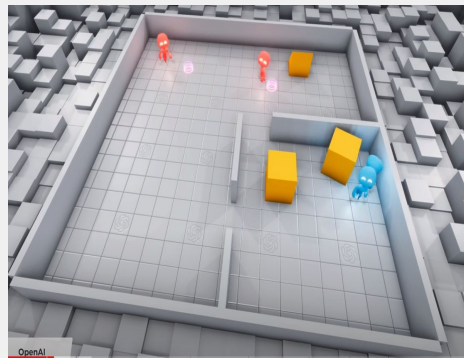


RL in practice

Trials



Agent



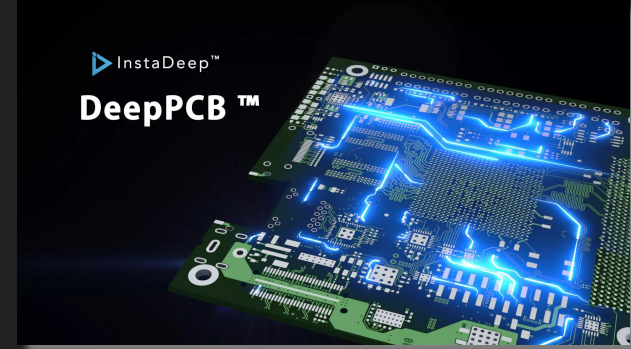
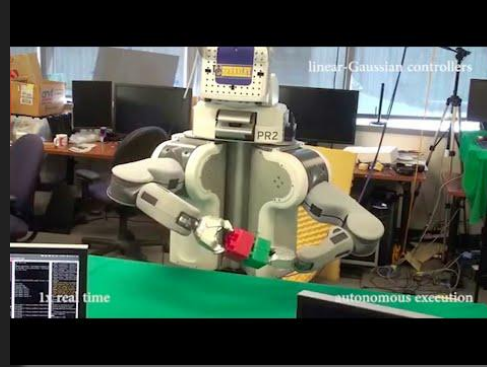
Environment

Learning

RL algorithms / Packages



Some applications



Drone racing champion



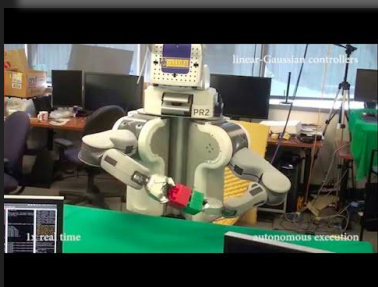
Article | [Open access](#) | Published: 05 October 2022

Discovering faster matrix multiplication algorithms with reinforcement learning

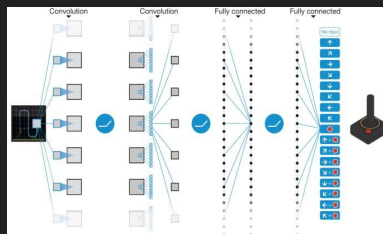
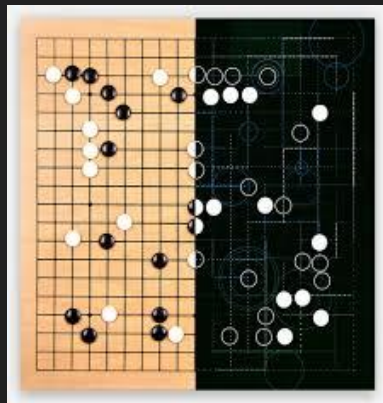
Not RL but fun to watch

Some examples

Robotics and Autonomous Systems



Games, Games, Games, and more Games



Some Other Industrial Applications

Chip Design with Deep Reinforcement Learning

Thursday, April 23, 2020

Posted by Anna Goldie, Senior Software Engineer and Azalia Mirhoseini, Senior Research Scientist, Google Research, Brain Team



20 JUL 2016

DeepMind AI Reduces Google Data Centre Cooling Bill by 40%

Article | [Open Access](#) | [Published: 16 February 2022](#)

Magnetic control of tokamak plasmas through deep reinforcement learning

[Not RL but fun to watch](#)

Why RL?

✓ Focus on **what** we want to achieve, instead of **how** we achieve.

Optimal policies are learned, not designed by domain-experts.

✓ (Mainly) **process-agnostic**; flexible to **adapt** to:

- New processes
- Operational constraints
- Optimization goals

Trading computation time for specialized personnel time.

✓ **No Heuristics**: learns from first principles, **unlocks efficiencies**.

✓ **Suited** for variability/perturbation handling.

✓ **Fast** and **scalable** approach. Results can be **competitive** with exact OR solvers but **10,000X faster**.

RL vs. SL & UL

Unsupervised Learning

Clusters or dimension reduction
k-means, PCA, etc.

Supervised Learning

Classifier or regressor
Neural networks, SVMs, etc.

Reinforcement Learning

Find an optimal behavior
Monte-Carlo, Q-learning, etc.

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Lots of new terms to be defined and connected to each other
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 - Anatomy of a RL solution
- Solution of a RL problem
Zoo of different methods

RL's basic ingredients

How do we make good decisions?!

What is the situation?!

State

What are the possible actions?

Action space

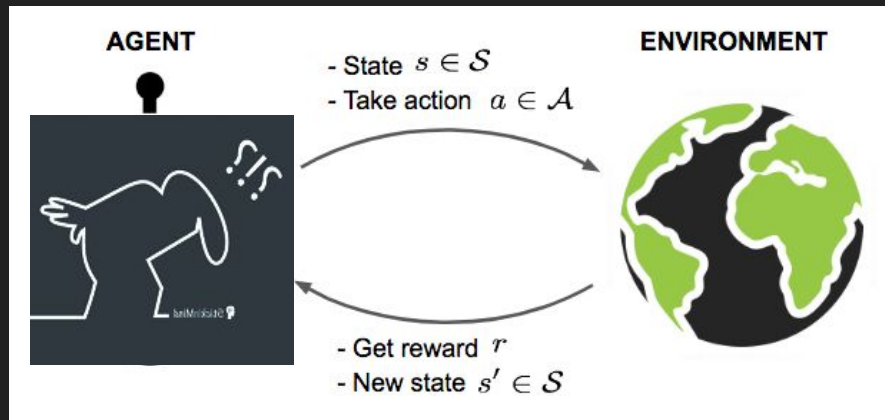
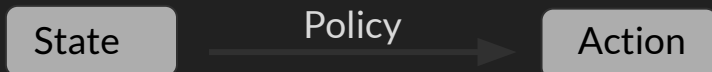
What are the consequences of each action?

- How rewarding/costly is each action?
REWARD
- Where do I end up after this action?

Environment

Policy

The mapping between the state and the actions



RL agent's job

Finding a **behavior** for **maximizing** the **sum of all rewards**, i.e. current reward and what comes after.

Examples

Give me examples!

- What are the states?
- What are the actions?
- What are the rewards at each step?
- What is your policy?

*What is RL
"Hello World!"?*

RL's problem formulation

Can you cast your decision making problem into a problem for the RL agent?

Reward hypothesis

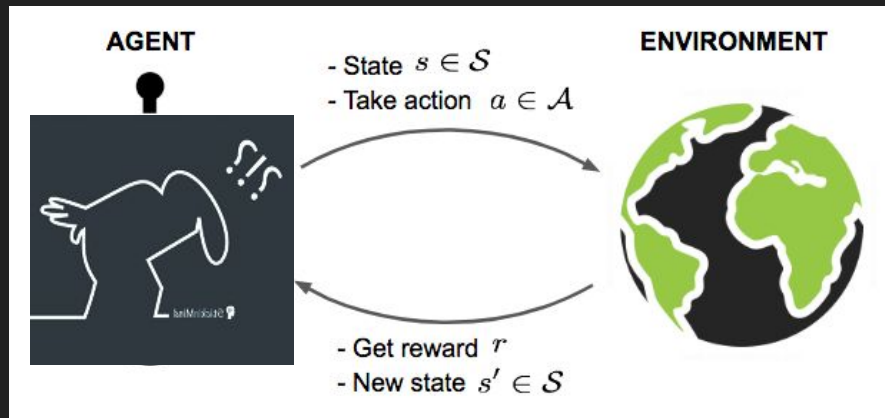
"All of what we mean by

goals and purposes can be well thought of as

maximization of the expected value of

the **cumulative sum** of

a received **scalar signal** (reward)." *Richard Sutton*



Reward mess up

- Can you think of a badly designed reward?

Anatomy of a RL solution

Find the optimal behavior

- Start with a policy
- Make it better (?!)
until you reach the optimal policy.

The questions will answer by the end of the course

- How good is a policy? (**policy evaluation**)
 - Use to policy in interaction with the env.
 - Measure certain quantities to evaluate your policy
- How can you make a policy better? (**policy improvement**)



Anatomy of a RL solution

Find the optimal behavior

- Start with a policy
- Make it better (!?)
until you reach the optimal policy.

The questions will answer by the end of the course

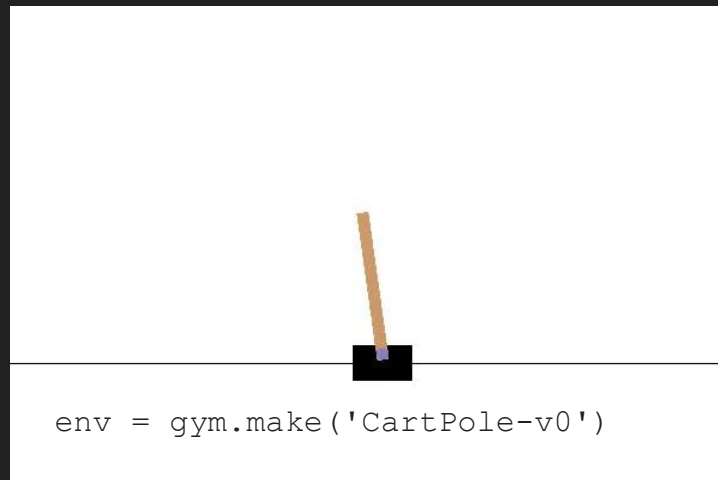
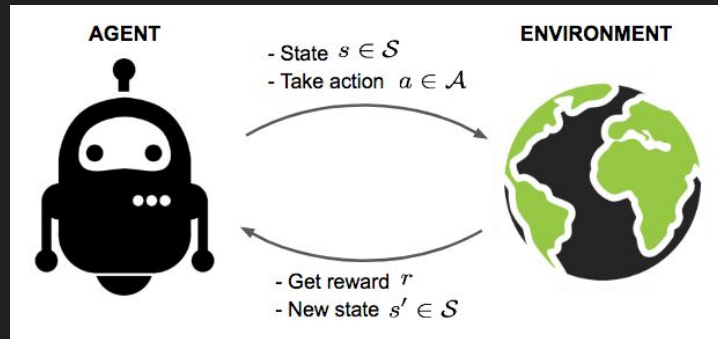
- How good is a policy? (policy evaluation)
 - Use to policy in interaction with the env.
 - Measure certain quantities to evaluate your policy
- How can you make a policy better? (policy improvement)

The rest of the course

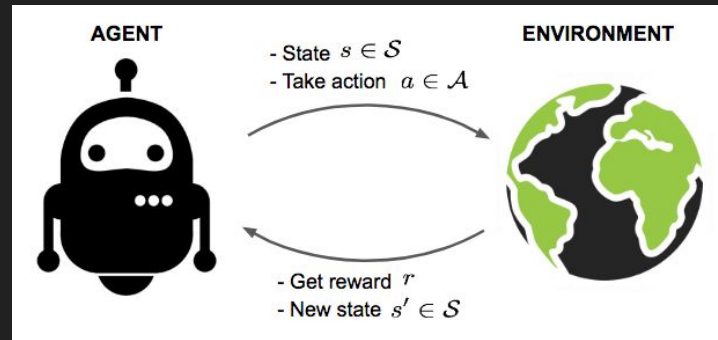
- Interaction with Environment
 - The interface between agent and environment
 - Where are the env.? How can I write my own?
- Your 1st Agent
 - Random agent
 - Learning agent
- Policy Evaluation
 - How to represent policy in a computer?
 - Measure of a good policy, Value function
 - Why values are important?
 - Ways to calculate those measures
 - Trial and error: Monte-Carlo
- Policy Improvement
 - A simple algorithm
 - Revisiting policy evaluation (SARSA)
 - Combining eval. and improv. (Q-learning)
- Introducing Deep Neural Nets
 - DQN: back to supervised learning

Interaction with Env.

- Let's find a *minimalistic* Agent <--> Environment *interface* together!
Gedankenexperiment: controlling from far far away!
- Where can I find an environment?
- When do I need to write my own?
- Properties of a good env.?
- Let's investigate an environment together!



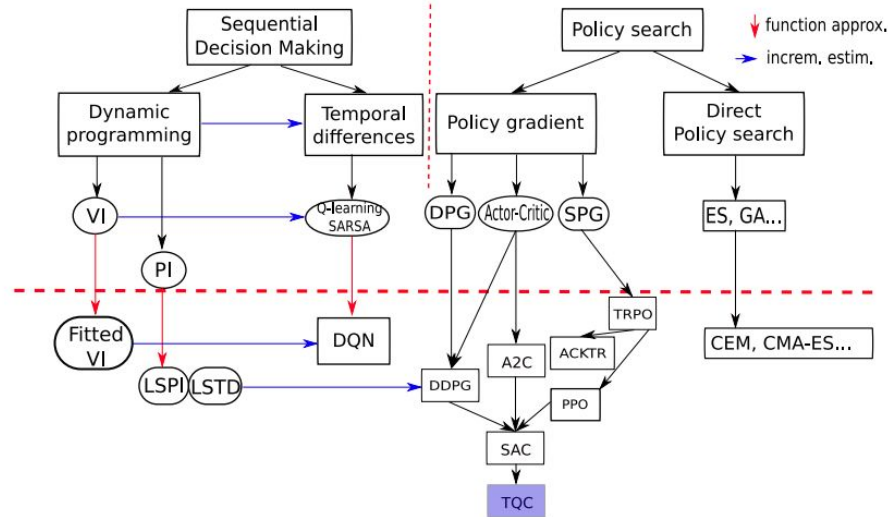
Your 1st agents



```
env = gym.make('CartPole-v0')
```

An overview of RL-algorithms taxonomy

The Big Picture

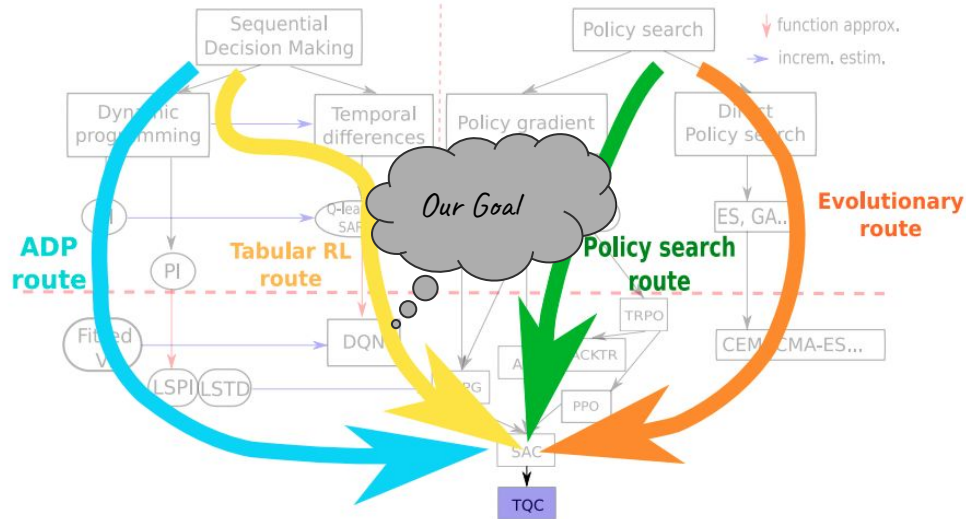


Olivier Sigaud

► A very partial view of the whole RL literature

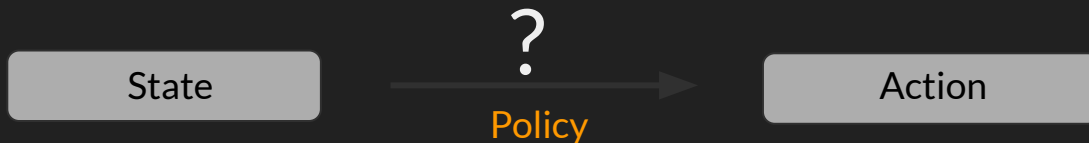
An overview of RL-algorithms taxonomy

The four routes to deep RL



- Four different ways to come to Deep RL

How to represent a policy?



Some possibilities

- Table,
- Function, or
- Deep neural network (*Deep RL*)

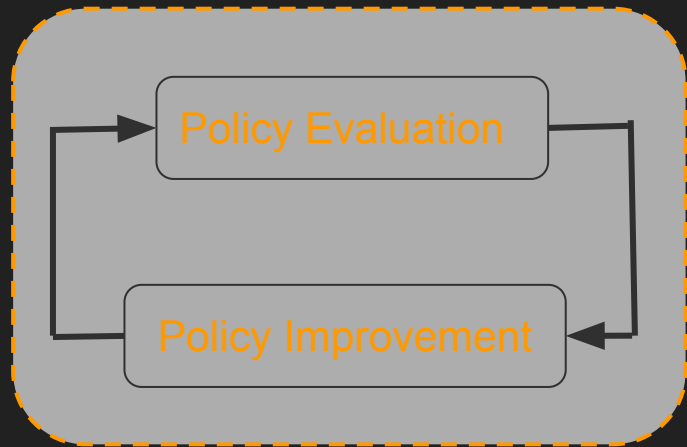
Different policy types

- Deterministic
- Stochastic

The core questions

- How good is a policy?
- How can you make a policy better?

Anatomy of a RL solution



My claim: The optimization problem is solved if
you can find how good your policy is

Policy Evaluation

Q: What are the measures of a good policy?

Let's say you are in a particular state and you are offered two policies, how do you choose?

- Immediate reward? Probably not a good idea.
- Some of all rewards you get? That seems more appropriate!
- Is future that important? Maybe, yes, Maybe no!

Let's formally define values function.

Q: How to find out $V(s)$ for all the states?

- Directly solving the Bellman Equation
- Guessing based on the experimenting [Monte Carlo method, Q-Learning, DQN]



(Finally) Bellman Equation

- What happens until eternity is what happens now plus what happens after that

Let's derive the Bellman equation together!

- **Assumptions:**

- A Markovian process (is it an important assumption?)
- The dynamics is known!
- States and actions are discrete!

$$\begin{aligned} v_{\pi}(s) &\doteq \mathbb{E}_{\pi}[G_t \mid S_t = s] \\ &= \mathbb{E}_{\pi}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r | s, a) \left[r + \gamma \mathbb{E}_{\pi}[G_{t+1} | S_{t+1} = s'] \right] \\ &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) \left[r + \gamma v_{\pi}(s') \right], \quad \text{for all } s \in \mathcal{S}, \end{aligned}$$

- Bellman equation has a good mathematical property! A lovely one!

What did we do?

Goal: finding the optimal behavior, without prior knowledge of the environment.

These are methods which require only **experience**!

Experience: sample sequences of states, actions, and rewards (from interaction with the environment).

Policy Evaluation Cntd.

Different policy types

- Deterministic
- Stochastic

Q1: How do you evaluate a deterministic policy?

Don't think of computational cost, we deal with that later.

Q2: What about a stochastic policy?

Q3: Why are stochastic policy important even for deterministic problem?

Policy Improvement

Q-values

Quality of an action:

How good is action a from state s , if I follow policy π after that action?

Policy improvement

At one state change the policy to $\operatorname{argmax} Q(s,a)$!

*Congratulations! You
arrived at
Monte-Carlo Control*

Q1: By the suggested policy improvement, the policy becomes deterministic. Can you suggest some way of making the policy better but staying stochastic?

How to solve the Bellman equation?

How to solve an equation?

- Direct analytical methods
- Numerical (iterative/approximate) methods: Not “*the*” solution but a *good enough* solution

When to use which?

- exists an analytical approach and if it is computationally feasible → analytical
- otherwise → numerical methods

How to solve the Bellman equation? (Cntd.)

$$x = f(x)$$

- Drawing (yes, why not?!)
- Midpoint method
- Iterative fixed-point methods
- FfT: Perturbative methods (start from where you know the best!)

How to solve the Bellman equation? (Cntd.)

Exercise

- Choose a policy
- Evaluate the policy
- (Use the evaluation to) Improve the policy
 - Unless you are happy(!) go back to Evaluation Step

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
 $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$
2. Policy Evaluation
Loop:
 $\Delta \leftarrow 0$
Loop for each $s \in \mathcal{S}$:
 $v \leftarrow V(s)$
 $V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s)) [r + \gamma V(s')]$
 $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)
3. Policy Improvement
 $policy_stable \leftarrow true$
For each $s \in \mathcal{S}$:
 $old_action \leftarrow \pi(s)$
 $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$
If $old_action \neq \pi(s)$, then $policy_stable \leftarrow false$
If $policy_stable$, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

What if?

- What if the states/actions do not fit into a table?
- What if the environment is not fully observable?
- What if the environment is stochastic?
- What if the optimal policy is stochastic?
- What if the process is not Markovian?
- What if we **do not know the dynamics** of the environment?

What **is** left for us?



Summary so far

- Basics of an RL problem
- The cornerstone of RL $\rightarrow$ Bellmann Eq.
- An iterative evaluation of a policy & Improving the policy to the optimal one



Monte Carlo



Algorithms relying on [repeated](#) random sampling!

Applications: any problem having a probabilistic *interpretation*.

Give me examples!

Examples:

- Finding a probability distribution of dice
- Finding the area of lake
- Finding the value function!

Monte Carlo Control

Monte-Carlo Control

- Choose a policy
- For many iterations:
 - Evaluate the Q values of this policy using MC
 - Improve the policy using the Qs
 - Make the new policy “soft”

Q-learning

How could we make the previous algorithm better?

- What was in-efficient?
- How is it different from the way we learn?
- When did it took so long?

Let's Bootstrap!

Let's make the best out of what we have learned!

SARSA method

Q-learning (Cntd.)

“SARSA” method for policy evaluation

lets derive SARSA together!

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_t \square + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t) \right].$$

We can use replace the MC policy evaluation with “SARSA” in the general scheme.

But we can also do better: **Q-learning!**

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_t \square + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right].$$

Q-learning (Cntd.)

Exercise

Q-learning (off-policy TD control) for estimating $\pi \approx \pi_*$

Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$

Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+, a \in \mathcal{A}(s)$, arbitrarily except that $Q(\text{terminal}, \cdot) = 0$

Loop for each episode:

 Initialize S

 Loop for each step of episode:

 Choose A from S using policy derived from Q (e.g., ε -greedy)

 Take action A , observe R, S'

$Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_a Q(S', a) - Q(S, A)]$

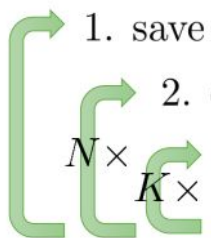
$S \leftarrow S'$

 until S is terminal

(D)DQN

Q-Learning with target networks

Q-learning with replay buffer and target network:

- 
1. save target network parameters: $\phi' \leftarrow \phi$
 2. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)\}$ using some policy, add it to $\mathcal{B}$
 3. sample a batch $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)$ from $\mathcal{B}$
 4. $\phi \leftarrow \phi - \alpha \sum_i \frac{dQ_\phi}{d\phi}(\mathbf{s}_i, \mathbf{a}_i) (Q_\phi(\mathbf{s}_i, \mathbf{a}_i) - \underbrace{[r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_{\phi'}(\mathbf{s}'_i, \mathbf{a}')]}_{\text{targets don't change in inner loop!}})$

supervised regression